

Generative AI, LLMs, Tokens, Embeddings & Prompt Engineering

Topic 1: What is Generative AI?

Generative AI is a type of Artificial Intelligence that creates new content.

Examples:

- Text
 - Images
 - Audio
 - Video
 - Code
-

Traditional AI

Input

↓

Prediction

Example:

Spam or Not Spam

Generative AI

Input Prompt

↓

Generate New Content

Example:

Write a leave application

Topic 2: What is an LLM?

LLM stands for:

Large Language Model

An LLM is a Deep Learning model trained on massive amounts of text data.

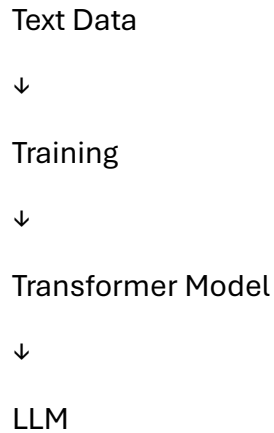
Examples:

- GPT
 - Claude
 - Gemini
 - Llama
 - Mistral
-

What Can LLMs Do?

- Answer questions
 - Generate code
 - Generate documents
 - Summarize content
 - Translate languages
 - Assist developers
-

LLM Architecture



Topic 3: What are Tokens?

LLMs do not understand words.

They understand:

Tokens

Example

Sentence:

I love Azure OpenAI

Tokens:

I

love

Azure

OpenAI

Or

I
love
Azure
Open
AI

depending on the tokenizer.

Why Tokens Matter?

Tokens determine:

- Cost
 - Input Length
 - Output Length
 - Context Window
-

Topic 4: Context Window

The context window is the amount of information an LLM can remember during a conversation.

Example:

Question 1
Question 2
Question 3

The model uses previous conversation context to answer.

Context Window Example

User:

My name is Bushi.

User:

What is my name?

ChatGPT remembers previous context.

Topic 5: What are Embeddings?

Embeddings convert text into numbers.

Example

Azure OpenAI

becomes

[0.234, 0.897, 1.234 ...]

Why Embeddings?

To measure similarity.

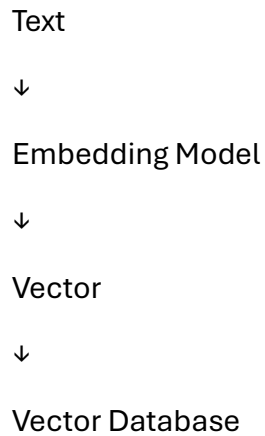
Example

These sentences have similar embeddings:

How to create Azure Service Bus?

How do I configure Service Bus?

Embedding Architecture



Topic 6: Vector Databases

Vector databases store embeddings.

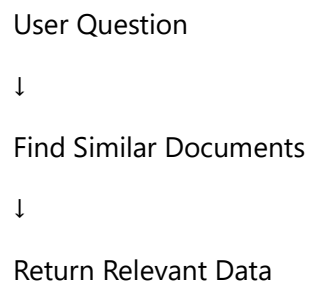
Examples:

- Azure AI Search
- Pinecone
- Weaviate
- ChromaDB

Why Vector Databases?

To find similar information.

Example:



Topic 7: Prompt Engineering

Prompt Engineering is the process of writing effective prompts.

Poor Prompt

Explain Azure.

Better Prompt

Explain Azure Service Bus for a Solution Architect with:

- Architecture
- Pros
- Cons
- Cost
- Alternatives

Prompt Formula

Role

+

Task

+

Context

+

Output Format

Example

Act as an Azure Solution Architect.

Explain Azure Service Bus.

Provide:

1. Architecture
2. Benefits

- 3. Limitations
- 4. Cost Considerations
- 5. Alternatives

Format as Markdown.

Topic 8: Temperature

Temperature controls creativity.

Low Temperature

0.1

Output:

More Accurate

Less Creative

High Temperature

1.0

Output:

More Creative

Less Predictable

Topic 9: How ChatGPT Generates Responses

User Prompt

↓

Tokenizer

↓

Embeddings

↓

Transformer

↓

Token Prediction

↓

Generated Response

Topic 10: Introduction to RAG

RAG means:

Retrieval Augmented Generation

Problem

LLM may not know company-specific data.

Solution

User Question



Vector Search



Relevant Documents



LLM



Answer

Topic 11: Introduction to AI Agents

AI Agents are LLM-powered systems that can:

- Reason
- Plan
- Execute actions
- Call tools

Examples:

- Copilot Agents
- Azure AI Agents
- Copilot Studio Agents